

DIAGNOSING HARDWARE FAILURES THEORY

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

ANS:

Common Hardware Issue	Typical Symptom
1. RAM failure	PC does not boot, blank screen, or repeated beep sounds.
2. Hard disk/SSD failure	Operating system does not load or shows disk error messages.
3. Power supply (SMPS) failure	PC does not turn on or suddenly shuts down.
4. CPU overheating	PC becomes very slow, freezes, or shuts down automatically.
5. Graphics card (GPU) problem	No display, screen artifacts, or display driver errors.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

ANS:

What to Check	What to Look For
1. Power socket	Check whether the wall socket is supplying power by testing another device.
2. Charger and cable	Check for cuts, loose connections, bent pins, or physical damage.
3. Charging port	Check the laptop charging port for dust, damage, or a loose connection.
4. Power adapter	Use a multimeter to check whether the charger is providing the correct output voltage.
5. Battery connection	If accessible, check whether the battery connector is properly connected and not damaged.
6. Test with a known-good charger	Try a compatible working charger to determine whether the original charger is faulty.
7. Final check	If it still does not charge, the problem may be with the battery, charging port, or motherboard charging circuit and may require professional repair.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

ANS:

USB Phone Charger Voltage Test

Test	Result
1.Expected USB voltage	4.75V – 5.25V
2.Measured voltage	5.0V
3.Conclusion	The measured voltage of 5.0V is within the expected range , so the charger output is normal.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

ANS:

ATX Power Supply 12V Rail Test

Steps:

1. Turn off the PC and **unplug it from the wall**.
2. Disconnect the ATX power connector from the motherboard.
3. Set the multimeter to **DC Voltage (V --)**.
4. Connect the **black probe to a black (ground) wire**.
5. Touch the **red probe to the yellow (+12V) wire**.
6. Record the voltage shown on the multimeter.

Measured Voltage: 12.0V

Result: 12.0V is within the normal range, so the 12V rail is working normally.

If voltage is below 11.5V: It could indicate a **faulty or weak SMPS/power supply**, which may cause random shutdowns.